1.42 Matrices: Invariant Points and Lines

Worked Solutions

1)

The origin, $(0, 0)$.

2 a)

$$\begin{pmatrix} 1 & 0 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

$\therefore (2, 5)$ is invariant under the transformation.

2 b)

$$\begin{pmatrix} 6 & -2 \\ 10 & -3 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

$\therefore (2, 5)$ is invariant under the transformation.

2 c)

$$\begin{pmatrix} 15 & -5 \\ 11 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

$\therefore (2, 5)$ is not invariant under the transformation.

2 d)

$$\begin{pmatrix} 21 & -8 \\ 100 & -39 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

$\therefore (2, 5)$ is invariant under the transformation.

2 e)

$$\begin{pmatrix} 50 & -21 \\ 31 & -12 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$$

$\therefore (2, 5)$ is not invariant under the transformation.

3 a)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$\therefore (1, -2, -1)$ is invariant under the transformation.

3 b)

$$\begin{pmatrix} 2 & -1 & 3 \\ 3 & 4 & -3 \\ 2 & -1 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$\therefore (1, -2, -1)$ is invariant under the transformation.

3 c)

$$\begin{pmatrix} 19 & 1 & 16 \\ 12 & 11 & -8 \\ -5 & 3 & -10 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$\therefore (1, -2, -1)$ is invariant under the transformation.

3 d)

$$\begin{pmatrix} \frac{1}{2} & -\frac{5}{12} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{4} & \frac{5}{6} \\ \frac{3}{2} & \frac{6}{5} & \frac{1}{10} \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{5}{3} \\ -1 \end{pmatrix}$$

$\therefore (1, -2, -1)$ is not invariant under the transformation.

4 a)

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} y \\ -x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$y = x$. $-x = y$. The only solution to $y = x$ and $y = -x$ is $x = 0$ and $y = 0$.

4 b)

$$\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{\sqrt{3}}{2}x - \frac{1}{2}y \\ \frac{1}{2}x + \frac{\sqrt{3}}{2}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\frac{\sqrt{3}}{2}x - \frac{1}{2}y = x \Rightarrow \sqrt{3}x - y = 2x \Rightarrow y = (\sqrt{3} - 2)x$$
$$\frac{1}{2}x + \frac{\sqrt{3}}{2}y = y \Rightarrow x + \sqrt{3}y = y \Rightarrow x = y - \sqrt{3}y$$
$$\Rightarrow y = \frac{1}{1 - \sqrt{3}}x \Rightarrow y = \frac{-1 - \sqrt{3}}{2}x$$

The only solution to $y = (\sqrt{3} - 2)x$ and $y = \frac{-1-\sqrt{3}}{2}x$ is $x = 0$ and $y = 0$.

4 c)

$$\begin{pmatrix} -1 & 5 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -x + 5y \\ -3x + 2y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-x + 5y = x \Rightarrow 2x - 5y = 0 \Rightarrow y = \frac{2}{5}x$. $-3x + 2y = y \Rightarrow 3x - y = 0 \Rightarrow y = 3x$. The only solution to $y = \frac{2}{5}x$ and $y = 3x$ is $x = 0$ and $y = 0$.

4 d)

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} x+y \\ y+z \\ x+z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$x + y = x \Rightarrow y = 0 \quad (1)$$
$$y + z = y \Rightarrow z = 0 \quad (2)$$
$$x + z = z \Rightarrow x = 0 \quad (3)$$

The only solution is $x = 0$, $y = 0$ and $z = 0$.

4 e)

$$\begin{pmatrix} -1 & 1 & 2 \\ -1 & 2 & 0 \\ 0 & 3 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} -x+y+2z \\ -x+2y \\ 3y-2z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$-x + y + 2z = x \Rightarrow -2x + y + 2z = 0 \quad (1)$$
$$-x + 2y = y \Rightarrow -x + y = 0 \quad (2)$$
$$3y - 2z = z \Rightarrow 3y - 3z = 0 \Rightarrow y - z = 0 \quad (3)$$

The only solution is $x = 0$, $y = 0$ and $z = 0$.

4 f)

$$\begin{pmatrix} 2 & 3 & -1 \\ -3 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} 2x+3y-z \\ -3x+y+2z \\ -2x+2y+3z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$2x + 3y - z = x \Rightarrow x + 3y - z = 0 \quad (1)$$
$$-3x + y + 2z = y \Rightarrow -3x + 2z = 0 \quad (2)$$
$$-2x + 2y + 3z = z \Rightarrow -2x + 2y + 2z = 0 \Rightarrow -x + y + z = 0 \quad (3)$$

The only solution is $x = 0$, $y = 0$ and $z = 0$.

5)

$$\begin{pmatrix} 9 & 2\sqrt{5} \\ 16 & 1+4\sqrt{5} \end{pmatrix} \begin{pmatrix} \sqrt{5} \\ -4 \end{pmatrix} = \begin{pmatrix} 9\sqrt{5} - 8\sqrt{5} \\ 16\sqrt{5} - 4(1+4\sqrt{5}) \end{pmatrix} = \begin{pmatrix} \sqrt{5} \\ 16\sqrt{5} - 4 - 16\sqrt{5} \end{pmatrix} = \begin{pmatrix} \sqrt{5} \\ -4 \end{pmatrix}$$

$\therefore (\sqrt{5}, -4)$ is an invariant point.

6)

$$\begin{pmatrix} \frac{3}{11} & \frac{10}{11} \\ \frac{3}{5} & \frac{1}{4} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{3}{11}x + \frac{10}{11}y \\ \frac{3}{5}x + \frac{1}{4}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\frac{3}{11}x + \frac{10}{11}y = x \Rightarrow -\frac{8}{11}x + \frac{10}{11}y = 0 \Rightarrow -8x + 10y = 0$$

$$\Rightarrow y = \frac{8}{10}x \Rightarrow y = \frac{4}{5}x$$

$$\frac{3}{5}x + \frac{1}{4}y = y \Rightarrow \frac{3}{5}x - \frac{3}{4}y = 0 \Rightarrow 12x - 15y = 0$$

$$\Rightarrow y = \frac{12}{15}x \Rightarrow y = \frac{4}{5}x$$

So another point that would be invariant is $(1, \frac{4}{5})$.

7)

$$\begin{pmatrix} -\frac{1}{8} & -\frac{27}{16} \\ -\frac{1}{4} & \frac{5}{8} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -\frac{1}{8}x - \frac{27}{16}y \\ -\frac{1}{4}x + \frac{5}{8}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$-\frac{1}{8}x - \frac{27}{16}y = x \Rightarrow \frac{9}{8}x + \frac{27}{16}y = 0 \Rightarrow 18x + 27y = 0 \Rightarrow y = -\frac{18}{27}x \Rightarrow y = -\frac{2}{3}x$$

$$-\frac{1}{4}x + \frac{5}{8}y = y \Rightarrow \frac{1}{4}x + \frac{3}{8}y = 0 \Rightarrow 2x + 3y = 0 \Rightarrow y = -\frac{2}{3}x$$

So another point that would be invariant is $(1, -\frac{2}{3})$.

8)

$$\begin{pmatrix} \frac{83}{19} & -\frac{24}{19} \\ -\frac{32}{19} & \frac{31}{19} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{83}{19}x - \frac{24}{19}y \\ -\frac{32}{19}x + \frac{31}{19}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\frac{83}{19}x - \frac{24}{19}y = x \Rightarrow \frac{64}{19}x - \frac{24}{19}y = 0 \Rightarrow 64x - 24y = 0 \Rightarrow y = \frac{64}{24}x \Rightarrow y = \frac{8}{3}x$$

$$-\frac{32}{19}x + \frac{31}{19}y = y \Rightarrow -\frac{32}{19}x + \frac{12}{19}y = 0 \Rightarrow -32x + 12y = 0 \Rightarrow y = \frac{32}{12}x \Rightarrow y = \frac{8}{3}x$$

So two points that satisfy both of these equations are $(0, 0)$ and $(1, \frac{8}{3})$.

9)

$$\begin{pmatrix} 2 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} 2x + 2z \\ x + 3y \\ 2y - z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$2x + 2z = x \Rightarrow x + 2z = 0 \Rightarrow x = -2z \quad (1)$$

$$x + 3y = y \Rightarrow x + 2y = 0 \Rightarrow x = -2y \quad (2)$$

$$2y - z = z \Rightarrow 2y - 2z = 0 \Rightarrow y = z \quad (3)$$

Let $z = 1$, then $x = -2$ and $y = 1$. So another point that would be invariant is $(-2, 1, 1)$.

10)

$$\begin{pmatrix} 1 & -4 & -6 \\ 4 & -3 & 2 \\ -3 & 2 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} x - 4y - 6z \\ 4x - 3y + 2z \\ -3x + 2y - 2z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$x - 4y - 6z = x \Rightarrow -4y - 6z = 0 \Rightarrow 2y + 3z = 0 \quad (1)$$

$$4x - 3y + 2z = y \Rightarrow 4x - 4y + 2z = 0 \Rightarrow 2x - 2y + z = 0 \quad (2)$$
$$-3x + 2y - 2z = z \Rightarrow -3x + 2y - 3z = 0 \quad (3)$$

Rearrange (1): $y = -\frac{3}{2}z$ and substitute into (2):

$$2x - 2\left(-\frac{3}{2}z\right) + z = 0 \Rightarrow 2x + 4z = 0 \Rightarrow x + 2z = 0 \Rightarrow x = -2z \quad (4)$$

Let $z = 2$, then $x = -4$ and $y = -3$. So another point that would be invariant is $(-4, -3, 2)$.

11)

$$\begin{pmatrix} \frac{10}{m+3} & \frac{m-7}{6m+18} \\ \frac{m+1}{m+3} & \frac{5m+17}{6m+18} \end{pmatrix} \begin{pmatrix} 1 \\ 6 \end{pmatrix} = \begin{pmatrix} \frac{10}{m+3} + \frac{6(m-7)}{6m+18} \\ \frac{m+1}{m+3} + \frac{6(5m+17)}{6m+18} \end{pmatrix}$$
$$= \begin{pmatrix} \frac{10}{m+3} + \frac{m-7}{m+3} \\ \frac{m+1}{m+3} + \frac{5m+17}{m+3} \end{pmatrix} = \begin{pmatrix} \frac{10+m-7}{m+3} \\ \frac{m+1+5m+17}{m+3} \end{pmatrix} = \begin{pmatrix} \frac{m+3}{m+3} \\ \frac{6m+18}{m+3} \end{pmatrix} = \begin{pmatrix} \frac{m+3}{m+3} \\ \frac{6(m+3)}{m+3} \end{pmatrix} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$$

$\therefore (1, 6)$ is invariant under the transformation.

12 a)

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -x \\ -y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-x = x \Rightarrow x = 0$. $-y = y \Rightarrow y = 0$. So $(0, 0)$ is the only invariant point. This is because the matrix represents a rotation by 180° about the origin.

12 b)

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$y = x$. $x = y \Rightarrow y = x$. So $y = x$ is a line of invariant points. This is because the matrix represents a reflection in the line $y = x$.

12 c)

$$\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} 4x \\ 4y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$4x = x \Rightarrow x = 0$. $4y = y \Rightarrow y = 0$. So $(0, 0)$ is the only invariant point. This is because the matrix represents an enlargement about the origin by scale factor 4.

12 d)

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -y \\ -x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-y = x \Rightarrow y = -x$. $-x = y \Rightarrow y = -x$. So $y = -x$ is a line of invariant points. This is because the matrix represents a reflection in the line $y = -x$.

12 e)

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-y = x \Rightarrow y = -x$. $x = y \Rightarrow y = x$. So $(0, 0)$ is the only invariant point. This is because the matrix represents a rotation by 90°, anticlockwise about the origin.

12 f)

$$\begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} x + 6y \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$x + 6y = x \Rightarrow 6y = 0 \Rightarrow y = 0$. $y = y$. So $y = 0$ is a line of invariant points. This is because the matrix represents a shear, with the x-axis fixed, with $(0, 1)$ mapped to $(6, 1)$.

12 g)

$$\begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ -4x + y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$x = x$. $-4x + y = y \Rightarrow -4x = 0 \Rightarrow x = 0$. So $x = 0$ is a line of invariant points. This is because the matrix represents a shear, with the y-axis fixed, with $(1, 0)$ mapped to $(1, -4)$.

12 h)

$$\begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} 4x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$4x = x \Rightarrow x = 0$. $y = y$. So $x = 0$ is a line of invariant points. This is because the matrix represents a stretch, parallel to the x-axis, scale factor 4.

12 i)

$$\begin{pmatrix} -2 & -2 \\ 3 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -2x - 2y \\ 3x + 3y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-2x - 2y = x \Rightarrow 3x + 2y = 0 \Rightarrow y = -\frac{3}{2}x$. $3x + 3y = y \Rightarrow 3x + 2y = 0 \Rightarrow y = -\frac{3}{2}x$. So $y = -\frac{3}{2}x$ is a line of invariant points.

12 j)

$$\begin{pmatrix} -3 & -3 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -3x - 3y \\ 2x + 2y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-3x - 3y = x \Rightarrow 4x + 3y = 0 \Rightarrow y = -\frac{4}{3}x$. $2x + 2y = y \Rightarrow 2x + y = 0 \Rightarrow y = -2x$. So $(0, 0)$ is the only invariant point.

13)

$$\begin{pmatrix} 8 & 4 \\ -3 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} 8x + 4y \\ -3x - 2y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$8x + 4y = x \Rightarrow 7x + 4y = 0 \Rightarrow y = -\frac{7}{4}x$. $-3x - 2y = y \Rightarrow 3x + 3y = 0 \Rightarrow y = -x$. $y = -\frac{7}{4}x$ and $y = -x$ intersect only at the origin, so there are no lines of invariant points.

14)

$$\begin{pmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{3}{5}x + \frac{4}{5}y \\ \frac{4}{5}x - \frac{3}{5}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\frac{3}{5}x + \frac{4}{5}y = x \Rightarrow -\frac{2}{5}x + \frac{4}{5}y = 0 \Rightarrow -2x + 4y = 0 \Rightarrow y = \frac{1}{2}x$$
$$\frac{4}{5}x - \frac{3}{5}y = y \Rightarrow -\frac{4}{5}x + \frac{8}{5}y = 0 \Rightarrow -4x + 8y = 0 \Rightarrow y = \frac{1}{2}x$$

As the transformation is a reflection, any points that lie on the mirror line must be invariant, and hence the mirror line itself is a line of invariant points. The equation of the mirror line is $y = \frac{1}{2}x$.

15)

$$\begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & \frac{k^2-1}{1+k^2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{1-k^2}{1+k^2}x + \frac{2k}{1+k^2}y \\ \frac{2k}{1+k^2}x + \frac{k^2-1}{1+k^2}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\frac{1-k^2}{1+k^2}x + \frac{2k}{1+k^2}y = x \Rightarrow (1-k^2)x + 2ky = (1+k^2)x$$
$$\Rightarrow x - k^2x + 2ky = x + k^2x \Rightarrow 2ky = 2k^2x \Rightarrow y = kx$$
$$\frac{2k}{1+k^2}x + \frac{k^2-1}{1+k^2}y = y \Rightarrow 2kx + (k^2-1)y = (1+k^2)y$$
$$\Rightarrow 2kx + k^2y - y = y + k^2y \Rightarrow 2kx = 2y \Rightarrow y = kx$$

$\therefore y = kx$ is a line of invariant points. Alternative Method:

$$\begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & \frac{k^2-1}{1+k^2} \end{pmatrix} \begin{pmatrix} x \\ kx \end{pmatrix} = \begin{pmatrix} \frac{1-k^2}{1+k^2}x + \frac{2k}{1+k^2} \times kx \\ \frac{2k}{1+k^2}x + \frac{k^2-1}{1+k^2} \times kx \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1-k^2}{1+k^2}x + \frac{2k^2}{1+k^2}x \\ \frac{2k}{1+k^2}x + \frac{k^3-k}{1+k^2}x \end{pmatrix} = \begin{pmatrix} \frac{1-k^2+2k^2}{1+k^2}x \\ \frac{2k+k^3-k}{1+k^2}x \end{pmatrix} = \begin{pmatrix} \frac{1+k^2}{1+k^2}x \\ \frac{k+k^3}{1+k^2}x \end{pmatrix} = \begin{pmatrix} x \\ \frac{(1+k^2)k}{1+k^2}x \end{pmatrix} = \begin{pmatrix} x \\ kx \end{pmatrix}$$

$\therefore y = kx$ is a line of invariant points.

16 a)

$$\begin{pmatrix} 1 & -1 & 2 \\ -1 & 0 & 3 \\ 3 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} x - y + 2z \\ -x + 3z \\ 3x - y \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$x - y + 2z = x \Rightarrow -y + 2z = 0 \Rightarrow y = 2z \quad (1)$$
$$-x + 3z = y \Rightarrow -x - y + 3z = 0 \quad (2)$$
$$3x - y = z \Rightarrow 3x - y - z = 0 \quad (3)$$

Substitute (1) into (2): $-x - (2z) + 3z = 0 \Rightarrow -x + z = 0 \Rightarrow x = z$ (4). So $2z = y$ and $2z = 2x$. The line of invariant points has Cartesian equation $2x = y = 2z$ or $\frac{x}{1/2} = y = \frac{z}{1/2}$.

16 b)

$$\begin{pmatrix} -1 & 0 & 4 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} -x + 4z \\ 2x + 3y + 2z \\ x + y + 2z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$-x + 4z = x \Rightarrow -2x + 4z = 0 \Rightarrow x = 2z \quad (1)$$
$$2x + 3y + 2z = y \Rightarrow 2x + 2y + 2z = 0 \Rightarrow x + y + z = 0 \quad (2)$$
$$x + y + 2z = z \Rightarrow x + y + z = 0 \quad (3)$$

Substitute (1) into (2): $(2z) + y + z = 0 \Rightarrow y + 3z = 0 \Rightarrow y = -3z$ (4). So $z = \frac{1}{2}x$ and $z = -\frac{1}{3}y$. The line of invariant points has Cartesian equation $\frac{1}{2}x = -\frac{1}{3}y = z$ or $\frac{x}{2} = \frac{y}{-3} = z$.

17 a)

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-x = x \Rightarrow x = 0$. $y = y \Rightarrow y \in \mathbb{R}$. The line $x = 0$ is a line of invariant points. Any line of the form $y = k$, where k is a constant, is an invariant line.

17 b)

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-y = x \Rightarrow y = -x$. $x = y \Rightarrow y = x$. These lines intersect at $(0, 0)$, and so the origin is the only invariant point.

17 c)

$$\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} 3x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$3x = x \Rightarrow x = 0$. $y = y \Rightarrow y \in \mathbb{R}$. The line $x = 0$ is a line of invariant points. Any line of the form $y = k$, where k is a constant, is an invariant line.

17 d)

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ -y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$x = x \Rightarrow x \in \mathbb{R}$. $-y = y \Rightarrow y = 0$. The line $y = 0$ is a line of invariant points. Any line of the form $x = k$, where k is a constant, is an invariant line.

17 e)

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -x \\ -y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-x = x \Rightarrow x = 0$. $-y = y \Rightarrow y = 0$. These lines intersect at $(0, 0)$, and so the origin is the only invariant point.

17 f)

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ 2y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$x = x \Rightarrow x \in \mathbb{R}$. $2y = y \Rightarrow y = 0$. The line $y = 0$ is a line of invariant points. Any line of the form $x = k$, where k is a constant, is an invariant line.

17 g)

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$y = x$. $x = y \Rightarrow y = x$. The line $y = x$ is a line of invariant points. Any line of the form $y = -x + k$, where k is a constant, is an invariant line.

17 h)

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -y \\ -x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-y = x \Rightarrow y = -x$. $-x = y \Rightarrow y = -x + c$. The line $y = -x$ is a line of invariant points. Any line of the form $y = x + k$, where k is a constant, is an invariant line.

18 a)

The x-axis is a line of invariant points.

18 b)

The plane $y = 0$ is a plane of invariant points. Any line perpendicular to $y = 0$ would be an invariant line, i.e. $\mathbf{r} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$.

19 a)

$$\begin{pmatrix} 3 & 4 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \times 3 + 4 \times 3 \\ 2 \times 3 + 5 \times 3 \end{pmatrix} = \begin{pmatrix} 21 \\ 21 \end{pmatrix} = 7 \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

$\therefore k = 7$.

19 b)

$y = x$ is an invariant line under the transformation.

20 a)

$$\begin{pmatrix} -2 & 9 \\ 3 & -8 \end{pmatrix} \begin{pmatrix} -4 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \times -4 + 9 \times 4 \\ 3 \times -4 - 8 \times 4 \end{pmatrix} = \begin{pmatrix} 44 \\ -44 \end{pmatrix} = -11 \begin{pmatrix} -4 \\ 4 \end{pmatrix}$$

$\therefore k = -11$.

20 b)

$y = -x$ is an invariant line under the transformation.

20 c)

$$\begin{pmatrix} -2 & 9 \\ 3 & -8 \end{pmatrix} \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \times 6 + 9 \times 2 \\ 3 \times 6 - 8 \times 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \end{pmatrix} = 1 \begin{pmatrix} 6 \\ 2 \end{pmatrix}$$

$\therefore p = 1$.

20 d)

$y = \frac{2}{6}x \Rightarrow y = \frac{1}{3}x$ is a line of invariant points under the transformation, as $(6, 2)$ maps to itself.

21)

$$\begin{pmatrix} 2 & 1 \\ -5 & 8 \end{pmatrix} \begin{pmatrix} x \\ 5x \end{pmatrix} = \begin{pmatrix} 2 \times x + 5 \times x \\ -5 \times x + 8 \times 5x \end{pmatrix} = \begin{pmatrix} 7x \\ 35x \end{pmatrix} = 7 \begin{pmatrix} x \\ 5x \end{pmatrix}$$

$\therefore y = 5x$ is an invariant line under the transformation.

22)

$$\begin{pmatrix} -3 & 2 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} x \\ -\frac{1}{2}x \end{pmatrix} = \begin{pmatrix} -3 \times x + 2 \times (-\frac{1}{2}x) \\ 5 \times x + 6 \times (-\frac{1}{2}x) \end{pmatrix} = \begin{pmatrix} -4x \\ 2x \end{pmatrix} = -4 \begin{pmatrix} x \\ -\frac{1}{2}x \end{pmatrix}$$

$\therefore y = -\frac{1}{2}x$ is an invariant line under the transformation.

23)

$$\begin{pmatrix} -7 & -4 \\ -4 & 9 \end{pmatrix} \begin{pmatrix} x \\ \frac{1}{2-\sqrt{5}}x \end{pmatrix} = \begin{pmatrix} -7 \times x - 4 \times \frac{1}{2-\sqrt{5}}x \\ -4x + 9 \times \frac{1}{2-\sqrt{5}}x \end{pmatrix} = \begin{pmatrix} -7x - \frac{4}{2-\sqrt{5}}x \\ -4x + \frac{9}{2-\sqrt{5}}x \end{pmatrix}$$

$$= \begin{pmatrix} \frac{-7(2-\sqrt{5})-4}{2-\sqrt{5}}x \\ \frac{-4(2-\sqrt{5})+9}{2-\sqrt{5}}x \end{pmatrix} = \begin{pmatrix} \frac{-14+7\sqrt{5}-4}{2-\sqrt{5}}x \\ \frac{-8+4\sqrt{5}+9}{2-\sqrt{5}}x \end{pmatrix} = \begin{pmatrix} \frac{-18+7\sqrt{5}}{2-\sqrt{5}}x \\ \frac{1+4\sqrt{5}}{2-\sqrt{5}}x \end{pmatrix}$$

$$= \begin{pmatrix} \frac{-18+7\sqrt{5}}{2-\sqrt{5}} \times \frac{2+\sqrt{5}}{2+\sqrt{5}}x \\ \frac{1+4\sqrt{5}}{2-\sqrt{5}}x \end{pmatrix}$$

$$= \begin{pmatrix} \frac{-36-18\sqrt{5}+14\sqrt{5}+35}{4-5}x \\ \frac{1+4\sqrt{5}}{2-\sqrt{5}}x \end{pmatrix}$$

$$= \begin{pmatrix} (-1-4\sqrt{5})x \\ \frac{1+4\sqrt{5}}{2-\sqrt{5}}x \end{pmatrix}$$

$$= \begin{pmatrix} (1+4\sqrt{5})x \\ \frac{1+4\sqrt{5}}{2-\sqrt{5}}x \end{pmatrix}$$

$$= (1+4\sqrt{5}) \begin{pmatrix} x \\ \frac{1}{2-\sqrt{5}}x \end{pmatrix}$$

$\therefore y = \frac{1}{2-\sqrt{5}}x$ is an invariant line under the transformation.

24)

$$\begin{pmatrix} e^4 & e^{-2} \\ e^2 & e^{-4} \end{pmatrix} \begin{pmatrix} x \\ \frac{1}{e^2}x \end{pmatrix} = \begin{pmatrix} e^4 \times x + e^{-2} \times (\frac{1}{e^2}x) \\ e^2 \times x + e^{-4} \times (\frac{1}{e^2}x) \end{pmatrix}$$

$$= \begin{pmatrix} e^4 x + e^{-4}x \\ e^2 x + e^{-6}x \end{pmatrix}$$

$$= \begin{pmatrix} (e^4 + e^{-4})x \\ (e^2 + e^{-6})x \end{pmatrix} = (e^4 + e^{-4}) \begin{pmatrix} x \\ \frac{e^2+e^{-6}}{e^4+e^{-4}}x \end{pmatrix} = (e^4 + e^{-4}) \begin{pmatrix} x \\ \frac{e^2+e^{-6}}{e^4+e^{-4}} \times \frac{e^2}{e^2}x \end{pmatrix}$$

$$= (e^4 + e^{-4}) \begin{pmatrix} x \\ \frac{e^4+e^{-4}}{e^4+e^{-4}} \times \frac{1}{e^2}x \end{pmatrix} = (e^4 + e^{-4}) \begin{pmatrix} x \\ \frac{1}{e^2}x \end{pmatrix}$$

$\therefore y = \frac{1}{e^2}x$ is an invariant line under the transformation.

25)

$$\begin{pmatrix} \ln 2 & \ln 8 \\ \ln 4 & \ln 2 \end{pmatrix} \begin{pmatrix} x \\ \frac{\sqrt{6}}{3}x \end{pmatrix} = \begin{pmatrix} \ln 2 & 3\ln 2 \\ 2\ln 2 & \ln 2 \end{pmatrix} \begin{pmatrix} x \\ \frac{\sqrt{6}}{3}x \end{pmatrix} = \ln 2 \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ \frac{\sqrt{6}}{3}x \end{pmatrix}$$

$$= \ln 2 \begin{pmatrix} 1 \times x + 3 \times \frac{\sqrt{6}}{3}x \\ 2 \times x + 1 \times \frac{\sqrt{6}}{3}x \end{pmatrix}$$

$$= \ln 2 \begin{pmatrix} x + \sqrt{6}x \\ 2x + \frac{\sqrt{6}}{3}x \end{pmatrix}$$

$$= \ln 2 \begin{pmatrix} (1 + \sqrt{6})x \\ (2 + \frac{\sqrt{6}}{3})x \end{pmatrix}$$

$$= (1 + \sqrt{6}) \ln 2 \begin{pmatrix} x \\ \frac{2+\frac{\sqrt{6}}{3}}{1+\sqrt{6}}x \end{pmatrix}$$

$$= (1 + \sqrt{6}) \ln 2 \begin{pmatrix} x \\ \frac{6+\sqrt{6}}{3+3\sqrt{6}}x \end{pmatrix}$$

$$= (1 + \sqrt{6}) \ln 2 \begin{pmatrix} x \\ \frac{6+\sqrt{6}}{3+3\sqrt{6}} \times \frac{3-3\sqrt{6}}{3-3\sqrt{6}}x \end{pmatrix} = (1 + \sqrt{6}) \ln 2 \begin{pmatrix} x \\ \frac{-15\sqrt{6}}{-45}x \end{pmatrix} = (1 + \sqrt{6}) \ln 2 \begin{pmatrix} x \\ \frac{\sqrt{6}}{3}x \end{pmatrix}$$

$\therefore y = \frac{\sqrt{6}}{3}x$ is an invariant line under the transformation.

26)

$$\begin{pmatrix} \sin k \cos k & \sin k \cos k \\ \sin k \cos k & -\sin k \cos k \end{pmatrix} \begin{pmatrix} x \\ \frac{1}{1+\sqrt{2}}x \end{pmatrix} = \begin{pmatrix} \sin k \cos k \times x + \sin k \cos k \times \frac{1}{1+\sqrt{2}}x \\ \sin k \cos k \times x + (-\sin k \cos k) \times \frac{1}{1+\sqrt{2}}x \end{pmatrix}$$

$$= \begin{pmatrix} \sin k \cos k(1 + \frac{1}{1+\sqrt{2}})x \\ \sin k \cos k(1 - \frac{1}{1+\sqrt{2}})x \end{pmatrix} = \sin k \cos k \begin{pmatrix} (1 + \frac{1}{1+\sqrt{2}})x \\ (1 - \frac{1}{1+\sqrt{2}})x \end{pmatrix}$$

$$= \sin k \cos k(1 + \frac{1}{1+\sqrt{2}}) \begin{pmatrix} x \\ \frac{1-\frac{1}{1+\sqrt{2}}}{1+\frac{1}{1+\sqrt{2}}}x \end{pmatrix}$$

$$= \sin k \cos k(1 + \frac{1}{1+\sqrt{2}}) \begin{pmatrix} x \\ \frac{1+\sqrt{2}-1}{1+\sqrt{2}+1}x \end{pmatrix}$$

$$= \sin k \cos k(1 + \frac{1}{1+\sqrt{2}}) \begin{pmatrix} x \\ \frac{\sqrt{2}}{2+\sqrt{2}}x \end{pmatrix}$$
$$= \sin k \cos k(1 + \frac{1}{1+\sqrt{2}}) \begin{pmatrix} x \\ \frac{1}{2}\frac{2}{\sqrt{2}+1}x \end{pmatrix}$$
$$= \sin k \cos k(1 + \frac{1}{1+\sqrt{2}}) \begin{pmatrix} x \\ \frac{1}{1+\sqrt{2}}x \end{pmatrix}$$

$\therefore y = \frac{1}{1+\sqrt{2}}x$ is an invariant line under the transformation.

27 a)

$$\begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} x+y \\ -2x+4y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = x + y \Rightarrow X = x + (mx + c) \Rightarrow X = (m+1)x + c$$
$$Y = -2x + 4y \Rightarrow Y = -2x + 4(mx + c) \Rightarrow Y = (4m-2)x + 4c$$

$Y = mX + c$:

$$(4m-2)x + 4c = m((m+1)x + c) + c$$
$$\Rightarrow 4mx - 2x + 4c = m^2x + mx + mc + c$$
$$\Rightarrow 0 = m^2x - 3mx + 2x + mc - 3c \Rightarrow 0 = (m^2 - 3m + 2)x + (m-3)c$$

Compare coefficients:

$$m^2 - 3m + 2 = 0 \Rightarrow (m-2)(m-1) = 0 \Rightarrow m = 2 \text{ or } m = 1$$

$(m-3)c = 0 \Rightarrow m = 3$ or $c = 0$. So either $m = 2$ and $c = 0$, or $m = 1$ and $c = 0$. $\therefore y = 2x$ and $y = x$ are invariant lines.

$$\begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

So neither are lines of invariant points.

27 b)

$$\begin{pmatrix} 6 & 2 \\ -4 & 15 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 6x+2y \\ -4x+15y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = 6x + 2y \Rightarrow X = 6x + 2(mx + c) \Rightarrow X = (2m+6)x + 2c$$
$$Y = -4x + 15y \Rightarrow Y = -4x + 15(mx + c) \Rightarrow Y = (15m-4)x + 15c$$

$Y = mX + c$:

$$(15m-4)x + 15c = m((2m+6)x + 2c) + c \Rightarrow 15mx - 4x + 15c = 2m^2x + 6mx + 2mc + c$$
$$\Rightarrow 0 = 2m^2x - 9mx + 4x + 2mc - 14c \Rightarrow 0 = (2m^2 - 9m + 4)x + (2m-14)c$$

Compare coefficients:

$$2m^2 - 9m + 4 = 0 \Rightarrow (2m-1)(m-4) = 0 \Rightarrow m = \frac{1}{2} \text{ or } m = 4$$

$(2m-14)c = 0 \Rightarrow m = 7$ or $c = 0$. So either $m = \frac{1}{2}$ and $c = 0$, or $m = 4$ and $c = 0$. $\therefore y = \frac{1}{2}x$ and $y = 4x$ are invariant lines.

$$\begin{pmatrix} 6 & 2 \\ -4 & 15 \end{pmatrix} \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 7 \\ \frac{7}{2} \end{pmatrix}, \qquad \begin{pmatrix} 6 & 2 \\ -4 & 15 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 14 \\ 56 \end{pmatrix}$$

So neither are lines of invariant points.

27 c)

$$\begin{pmatrix} 5 & -3 \\ -1 & 7 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 5x-3y \\ -x+7y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = 5x - 3y \Rightarrow X = 5x - 3(mx + c) \Rightarrow X = (-3m + 5)x - 3c$$
$$Y = -x + 7y \Rightarrow Y = -x + 7(mx + c) \Rightarrow Y = (7m - 1)x + 7c$$

$Y = mX + c$:

$$(7m - 1)x + 7c = m((-3m + 5)x - 3c) + c \Rightarrow 7mx - x + 7c = -3m^2x + 5mx - 3mc + c$$

$$\Rightarrow 3m^2x + 2mx - x + 3mc + 6c = 0 \Rightarrow (3m^2 + 2m - 1)x + (3m + 6)c = 0$$

Compare coefficients:

$$3m^2 + 2m - 1 = 0 \Rightarrow (3m - 1)(m + 1) = 0 \Rightarrow m = \frac{1}{3} \text{ or } m = -1$$

$(3m + 6)c = 0 \Rightarrow m = -2$ or $c = 0$. So either $m = \frac{1}{3}$ and $c = 0$, or $m = -1$ and $c = 0$. $\therefore y = \frac{1}{3}x$ and $y = -x$ are invariant lines.

$$\begin{pmatrix} 5 & -3 \\ -1 & 7 \end{pmatrix}\begin{pmatrix} 1 \\ \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 4 \\ \frac{4}{3} \end{pmatrix}, \qquad \begin{pmatrix} 5 & -3 \\ -1 & 7 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 8 \\ -8 \end{pmatrix}$$

So neither are lines of invariant points.

27 d)

$$\begin{pmatrix} -25 & 24 \\ 12 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} -25x+24y \\ 12x+3y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = -25x + 24y \Rightarrow X = -25x + 24(mx + c) \Rightarrow X = (24m - 25)x + 24c$$
$$Y = 12x + 3y \Rightarrow Y = 12x + 3(mx + c) \Rightarrow Y = (3m + 12)x + 3c$$

$Y = mX + c$:

$$(3m + 12)x + 3c = m((24m - 25)x + 24c) + c$$
$$\Rightarrow 3mx + 12x + 3c = 24m^2x - 25mx + 24mc + c \Rightarrow 0 = 24m^2x - 28mx - 12x + 24mc - 2c$$
$$\Rightarrow 0 = 12m^2x - 14mx - 6x + 12mc - c \Rightarrow 2(6m^2 - 7m - 3)x + (12m - 1)c = 0$$

Compare coefficients:

$$6m^2 - 7m - 3 = 0 \Rightarrow (3m + 1)(2m - 3) = 0 \Rightarrow m = -\frac{1}{3} \text{ or } m = \frac{3}{2}$$

$(12m - 1)c = 0 \Rightarrow m = \frac{1}{12}$ or $c = 0$. So either $m = -\frac{1}{3}$ and $c = 0$, or $m = \frac{3}{2}$ and $c = 0$. $\therefore y = -\frac{1}{3}x$ and $y = \frac{3}{2}x$ are invariant lines.

$$\begin{pmatrix} -25 & 24 \\ 12 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} -33 \\ 11 \end{pmatrix}, \qquad \begin{pmatrix} -25 & 24 \\ 12 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 11 \\ \frac{33}{2} \end{pmatrix}$$

so neither are lines of invariant points.

27 e)

$$\begin{pmatrix} \frac{3}{7} & -\frac{24}{7} \\ -\frac{4}{7} & -\frac{17}{7} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{3}{7}x - \frac{24}{7}y \\ -\frac{4}{7}x - \frac{17}{7}y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = \frac{3}{7}x - \frac{24}{7}y \Rightarrow X = \frac{3}{7}x - \frac{24}{7}(mx + c) \Rightarrow X = (-\frac{24}{7}m + \frac{3}{7})x - \frac{24}{7}c$$
$$Y = -\frac{4}{7}x - \frac{17}{7}y \Rightarrow Y = -\frac{4}{7}x - \frac{17}{7}(mx + c) \Rightarrow Y = (-\frac{17}{7}m - \frac{4}{7})x - \frac{17}{7}c$$

$Y = mX + c$:

$$(-\frac{17}{7}m - \frac{4}{7})x - \frac{17}{7}c = m((-\frac{24}{7}m + \frac{3}{7})x - \frac{24}{7}c) + c$$

$$\Rightarrow -17mx - 4x - 17c = -24m^2x + 3mx - 24mc + 7c \Rightarrow 24m^2x - 20mx - 4x + 24mc - 24c = 0$$

$$\Rightarrow 6m^2x - 5mx - x + 6mc - 6c = 0 \Rightarrow (6m^2 - 5m - 1)x + (6m - 6)c = 0$$

Compare coefficients:

$$6m^2 - 5m - 1 = 0 \Rightarrow (6m + 1)(m - 1) = 0 \Rightarrow m = -\frac{1}{6} \text{ or } m = 1$$

$(6m - 6)c = 0 \Rightarrow m = 1$ or $c = 0$. So either $m = 1$ and c is any real value, or $m = -\frac{1}{6}$ and $c = 0$.
$\therefore y = x + c$ and $y = -\frac{1}{6}x$ are invariant lines.

$$\begin{pmatrix} \frac{3}{7} & -\frac{24}{7} \\ -\frac{4}{7} & -\frac{17}{7} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$$

so $y = x + c$ is an invariant line, but

$$\begin{pmatrix} \frac{3}{7} & -\frac{24}{7} \\ -\frac{4}{7} & -\frac{17}{7} \end{pmatrix} \begin{pmatrix} 1 \\ -\frac{1}{6} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{1}{6} \end{pmatrix}$$

so $y = -\frac{1}{6}x$ is a line of invariant points.

27 f)

$$\begin{pmatrix} \frac{39}{19} & -\frac{5}{19} \\ \frac{4}{19} & \frac{18}{19} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{39}{19}x - \frac{5}{19}y \\ \frac{4}{19}x + \frac{18}{19}y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = \frac{39}{19}x - \frac{5}{19}y \Rightarrow X = \frac{39}{19}x - \frac{5}{19}(mx + c) \Rightarrow X = (-\frac{5}{19}m + \frac{39}{19})x - \frac{5}{19}c$$
$$Y = \frac{4}{19}x + \frac{18}{19}y \Rightarrow Y = \frac{4}{19}x + \frac{18}{19}(mx + c) \Rightarrow Y = (\frac{18}{19}m + \frac{4}{19})x + \frac{18}{19}c$$

$Y = mX + c$:

$$(\frac{18}{19}m + \frac{4}{19})x + \frac{18}{19}c = m((-\frac{5}{19}m + \frac{39}{19})x - \frac{5}{19}c) + c$$

$$\Rightarrow 18mx + 4x + 18c = -5m^2x + 39mx - 5mc + 19c \Rightarrow 5m^2x - 21mx + 4x + 5mc - c = 0$$

$$\Rightarrow (5m^2 - 21m + 4)x + (5m - 1)c = 0$$

Compare coefficients:

$$5m^2 - 21m + 4 = 0 \Rightarrow (5m - 1)(m - 4) = 0 \Rightarrow m = \frac{1}{5} \text{ or } m = 4$$

$(5m-1)c=0 \Rightarrow m=\frac{1}{5}$ or $c=0$. So either $m=\frac{1}{5}$ and c is any real value, or $m=4$ and $c=0$.
$\therefore y=\frac{1}{5}x+c$ and $y=4x$ are invariant lines.

$$\begin{pmatrix} \frac{39}{19} & -\frac{5}{19} \\ \frac{4}{19} & \frac{18}{19} \end{pmatrix}\begin{pmatrix} 1 \\ \frac{1}{5} \end{pmatrix}=\begin{pmatrix} 2 \\ \frac{2}{5} \end{pmatrix}$$

so $y=\frac{1}{5}x+c$ is an invariant line, but

$$\begin{pmatrix} \frac{39}{19} & -\frac{5}{19} \\ \frac{4}{19} & \frac{18}{19} \end{pmatrix}\begin{pmatrix} 1 \\ 4 \end{pmatrix}=\begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

so $y=4x$ is a line of invariant points.

28)

$$\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 2x \\ x+y \end{pmatrix}=\begin{pmatrix} X \\ Y \end{pmatrix}$$

$X=2x$.

$$Y=x+y \Rightarrow Y=x+(mx+c) \Rightarrow Y=(m+1)x+c$$

$Y=mX+c$:

$$(m+1)x+c=m(2x)+c \Rightarrow mx+x=2mx \Rightarrow 0=mx-x \Rightarrow 0=(m-1)x$$

Compare coefficients: $m-1=0 \Rightarrow m=1$; $x=0$. So $m=1$ and c can be any real value, or $x=0$.
$\therefore y=x+c$ and $x=0$ are invariant lines.

$$\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}=\begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

so $y=x+c$ is an invariant line, but

$$\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix}=\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

so $x=0$ is a line of invariant points.

29)

$$\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 3x+y \\ y \end{pmatrix}=\begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X=3x+y \Rightarrow X=3x+(mx+c) \Rightarrow X=(3+m)x+c$$
$$Y=y \Rightarrow Y=mx+c$$

$Y=mX+c$:

$$mx+c=m((3+m)x+c)+c \Rightarrow mx+c=3mx+m^2x+mc+c$$

$$\Rightarrow 0=2mx+m^2x+mc \Rightarrow 0=(2m+m^2)x+mc$$

Compare coefficients:

$$2m+m^2=0 \Rightarrow m(2+m)=0 \Rightarrow m=0 \text{ or } m=-2$$

$mc = 0 \Rightarrow m = 0$ or $c = 0$. So $m = 0$ and c can be any real value, or $m = -2$ and $c = 0$. $\therefore y = c$ and $y = -2x$ are invariant lines.

$$\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

so $y = c$ is an invariant line, but

$$\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

so $y = -2x$ is a line of invariant points.

30)

$$\begin{pmatrix} 2 & -\frac{1}{2} \\ 8 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 2x - \frac{1}{2}y \\ 8x - 3y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = 2x - \frac{1}{2}y \Rightarrow X = 2x - \frac{1}{2}(mx + c) \Rightarrow X = (-\frac{1}{2}m + 2)x - \frac{1}{2}c$$
$$Y = 8x - 3y \Rightarrow Y = 8x - 3(mx + c) \Rightarrow Y = (-3m + 8)x - 3c$$

$Y = mX + c$:

$$(-3m + 8)x - 3c = m((-\frac{1}{2}m + 2)x - \frac{1}{2}c) + c$$
$$\Rightarrow -3mx + 8x - 3c = -\frac{1}{2}m^2x + 2mx - \frac{1}{2}mc + c$$
$$\Rightarrow \frac{1}{2}m^2x - 5mx + 8x + \frac{1}{2}mc - 4c = 0$$
$$\Rightarrow m^2x - 10mx + 16x + mc - 8c = 0 \Rightarrow (m^2 - 10m + 16)x + (m - 8)c = 0$$

Compare coefficients:

$$m^2 - 10m + 16 = 0 \Rightarrow (m - 8)(m - 2) = 0 \Rightarrow m = 8 \text{ or } m = 2$$

$(m - 8)c = 0 \Rightarrow m = 8$ or $c = 0$. So either $m = 8$ and c can be any real value, or $m = 2$ and $c = 0$. $\therefore y = 8x + c$ and $y = 2x$ are invariant lines.

$$\begin{pmatrix} 2 & -\frac{1}{2} \\ 8 & -3 \end{pmatrix} \begin{pmatrix} 1 \\ 8 \end{pmatrix} = \begin{pmatrix} -2 \\ -16 \end{pmatrix}$$

so $y = 8x + c$ is an invariant line, but

$$\begin{pmatrix} 2 & -\frac{1}{2} \\ 8 & -3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

so $y = 2x$ is a line of invariant points. So $p = 2$ and $q = 8$.

31)

$$\begin{pmatrix} -6 & -3 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} -6x - 3y \\ 5x - y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = -6x - 3y \Rightarrow X = -6x - 3(mx + c) \Rightarrow X = (-3m - 6)x - 3c$$
$$Y = 5x - y \Rightarrow Y = 5x - (mx + c) \Rightarrow Y = (-m + 5)x - c$$

$Y = mX + c$:

$$(-m+5)x - c = m((-3m-6)x - 3c) + c$$
$$\Rightarrow -mx + 5x - c = -3m^2x - 6mx - 3mc + c \Rightarrow 3m^2x + 5mx + 5x + 3mc - 2c = 0$$
$$\Rightarrow (3m^2 + 5m + 5)x + (3m-2)c = 0$$

Compare coefficients: $3m^2 + 5m + 5 = 0$.

$$b^2 - 4ac = (5)^2 - 4(3)(5) = -35 < 0$$

so no real roots. $(3m-2)c = 0 \Rightarrow m = \frac{2}{3}$ or $c = 0$. So there are no invariant lines.

32)

$$\begin{pmatrix} -10 & 6 \\ -8 & -15 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} -10x + 6y \\ -8x - 15y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = -10x + 6y \Rightarrow X = -10x + 6(mx + c) \Rightarrow X = (6m - 10)x + 6c$$
$$Y = -8x - 15y \Rightarrow Y = -8x - 15(mx + c) \Rightarrow Y = (-15m - 8)x - 15c$$

$Y = mX + c$:

$$(-15m - 8)x - 15c = m((6m - 10)x + 6c) + c$$
$$\Rightarrow -15mx - 8x - 15c = 6m^2x - 10mx + 6mc + c$$
$$\Rightarrow 0 = 6m^2x + 5mx + 8x + 6mc + 16c \Rightarrow 0 = (6m^2 + 5m + 8)x + (6m + 16)c$$

Compare coefficients: $6m^2 + 5m + 8 = 0$.

$$b^2 - 4ac = (5)^2 - 4(6)(8) = -167 < 0$$

so no real roots. $(6m + 16)c = 0 \Rightarrow m = -\frac{16}{6} = -\frac{8}{3}$ or $c = 0$. So there are no invariant lines.

33 a)

$$\begin{pmatrix} \frac{24}{25} & \frac{7}{25} \\ \frac{7}{25} & -\frac{24}{25} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{24}{25}x + \frac{7}{25}y \\ \frac{7}{25}x - \frac{24}{25}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\frac{24}{25}x + \frac{7}{25}y = x \Rightarrow -\frac{1}{25}x + \frac{7}{25}y = 0 \Rightarrow -x + 7y = 0 \Rightarrow y = \frac{1}{7}x$$
$$\frac{7}{25}x - \frac{24}{25}y = y \Rightarrow \frac{7}{25}x - \frac{49}{25}y = 0 \Rightarrow 7x - 49y = 0 \Rightarrow y = \frac{1}{7}x$$

$\therefore y = \frac{1}{7}x$ is a line of invariant points.

33 b)

$k = -7$ as the line must be perpendicular to the mirror line. The full worked solution is shown here:

$$\begin{pmatrix} \frac{24}{25} & \frac{7}{25} \\ \frac{7}{25} & -\frac{24}{25} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{24}{25}x + \frac{7}{25}y \\ \frac{7}{25}x - \frac{24}{25}y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = \frac{24}{25}x + \frac{7}{25}y \Rightarrow X = \frac{24}{25}x + \frac{7}{25}(mx + c) \Rightarrow X = (\frac{7}{25}m + \frac{24}{25})x + \frac{7}{25}c$$
$$Y = \frac{7}{25}x - \frac{24}{25}y \Rightarrow Y = \frac{7}{25}x - \frac{24}{25}(mx + c) \Rightarrow Y = (-\frac{24}{25}m + \frac{7}{25})x - \frac{24}{25}c$$

$Y = mX + c$:

$$(-\frac{24}{25}m + \frac{7}{25})x - \frac{24}{25}c = m((\frac{7}{25}m + \frac{24}{25})x + \frac{7}{25}c) + c$$
$$\Rightarrow -24mx + 7x - 24c = 7m^2x + 24mx + 7mc + 25c$$
$$\Rightarrow 0 = 7m^2x + 48mx - 7x + 7mc + 49c \Rightarrow 0 = (7m^2 + 48m - 7)x + (7m + 49)c$$

Compare coefficients:

$$7m^2 + 48m - 7 = 0 \Rightarrow (7m - 1)(m + 7) = 0 \Rightarrow m = \frac{1}{7} \text{ or } m = -7$$

$(7m + 49)c = 0 \Rightarrow m = -7$ or $c = 0$. So either $m = -7$ and c can be any real value, or $m = \frac{1}{7}$ and $c = 0$. $\therefore y = -7x + c$ and $y = \frac{1}{7}x$ are invariant lines.

$$\begin{pmatrix} \frac{24}{25} & \frac{7}{25} \\ \frac{7}{25} & -\frac{24}{25} \end{pmatrix} \begin{pmatrix} 1 \\ -7 \end{pmatrix} = \begin{pmatrix} -1 \\ 7 \end{pmatrix}$$

so $y = -7x + c$ is an invariant line, but

$$\begin{pmatrix} \frac{24}{25} & \frac{7}{25} \\ \frac{7}{25} & -\frac{24}{25} \end{pmatrix} \begin{pmatrix} 1 \\ \frac{1}{7} \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1}{7} \end{pmatrix}$$

so $y = \frac{1}{7}x$ is a line of invariant points.

34 a)

$$\begin{pmatrix} 28 & -9 \\ 4 & 16 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 28x - 9y \\ 4x + 16y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = 28x - 9y \Rightarrow X = 28x - 9(mx + c) \Rightarrow X = (-9m + 28)x - 9c$$
$$Y = 4x + 16y \Rightarrow Y = 4x + 16(mx + c) \Rightarrow Y = (16m + 4)x + 16c$$

$Y = mX + c$:

$$(16m + 4)x + 16c = m((-9m + 28)x - 9c) + c$$
$$\Rightarrow 16mx + 4x + 16c = -9m^2x + 28mx - 9mc + c$$
$$\Rightarrow 9m^2x - 12mx + 4x + 9mc + 15c = 0 \Rightarrow 0 = (9m^2 - 12m + 4)x + (9m + 15)c$$

Compare coefficients:

$$9m^2 - 12m + 4 = 0 \Rightarrow (3m - 2)^2 = 0 \Rightarrow m = \frac{2}{3}$$

$(9m + 15)c = 0 \Rightarrow m = -\frac{15}{9} = -\frac{5}{3}$ or $c = 0$. So $m = \frac{2}{3}$ and $c = 0$. $\therefore y = \frac{2}{3}x$ is an invariant line, i.e. $k = \frac{2}{3}$.

34 b)

$$\begin{pmatrix} 28 & -9 \\ 4 & 16 \end{pmatrix} \begin{pmatrix} 1 \\ \frac{2}{3} \end{pmatrix} = \begin{pmatrix} 22 \\ \frac{44}{3} \end{pmatrix}$$

so $y = \frac{2}{3}x$ is not a line of invariant points.

35)

$$\begin{pmatrix} 241 & -60 \\ 980 & -244 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 241x - 60y \\ 980x - 244y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = 241x - 60y \Rightarrow X = 241x - 60(mx + c) \Rightarrow X = (-60m + 241)x - 60c$$
$$Y = 980x - 244y \Rightarrow Y = 980x - 244(mx + c) \Rightarrow Y = (-244m + 980)x - 244c$$

$Y = mX + c$:

$$(-244m + 980)x - 244c = m((-60m + 241)x - 60c) + c$$
$$\Rightarrow -244mx + 980x - 244c = -60m^2x + 241mx - 60mc + c$$
$$\Rightarrow 60m^2x - 485mx + 980x + 60mc - 245c = 0$$
$$\Rightarrow 12m^2x - 97mx + 196x + 12mc - 49c = 0 \Rightarrow 0 = (12m^2 - 97m + 196)x + (12m - 49)c$$

Compare coefficients:

$$12m^2 - 97m + 196 = 0 \Rightarrow (12m - 49)(m - 4) = 0 \Rightarrow m = \frac{49}{12} \text{ or } m = 4$$

$(12m - 49)c = 0 \Rightarrow m = \frac{49}{12}$ or $c = 0$. So $m = \frac{49}{12}$ and $c \in \mathbb{R}$, or $m = 4$ and $c = 0$. $\therefore y = \frac{49}{12}x + c$ and $y = 4x$ are invariant lines.

$$\begin{pmatrix} 241 & -60 \\ 980 & -244 \end{pmatrix}\begin{pmatrix} 1 \\ \frac{49}{12} \end{pmatrix} = \begin{pmatrix} -4 \\ -\frac{49}{3} \end{pmatrix}$$

so $y = \frac{49}{12}x + c$ are not lines of invariant points.

$$\begin{pmatrix} 241 & -60 \\ 980 & -244 \end{pmatrix}\begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

so $y = 4x$ is a line of invariant points.

36)

$$\begin{pmatrix} 2 & k \\ -3 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 2x + ky \\ -3x + 4y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = 2x + ky \Rightarrow X = 2x + k(mx) \Rightarrow X = (km + 2)x$$
$$Y = -3x + 4y \Rightarrow Y = -3x + 4(mx) \Rightarrow Y = (4m - 3)x$$

$Y = mX + c$:

$$(4m - 3)x = m((km + 2)x) \Rightarrow 4mx - 3x = km^2x + 2mx$$
$$\Rightarrow 0 = km^2x - 2mx + 3x \Rightarrow 0 = (km^2 - 2m + 3)x$$

Compare coefficients: $km^2 - 2m + 3 = 0$. No invariant lines $\Rightarrow b^2 - 4ac < 0$.

$$\therefore (-2)^2 - 4(k)(3) < 0 \Rightarrow 4 - 12k < 0 \Rightarrow k > \frac{1}{3}$$

So if $k > \frac{1}{3}$ then $\mathbf{A}$ has no invariant lines that pass through the origin.

37)

$$\begin{pmatrix} 2k & 3 \\ -1 & 2k-3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 2kx+3y \\ -x+(2k-3)y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = 2kx + 3y \Rightarrow X = 2kx + 3(mx) \Rightarrow X = (2k+3m)x$$
$$Y = -x + (2k-3)y \Rightarrow Y = -x + (2k-3)(mx) \Rightarrow Y = (2km-3m-1)x$$

$Y = mX$:

$$(2km-3m-1)x = m((2k+3m)x) \Rightarrow 2kmx - 3mx - x = 2kmx + 3m^2x$$

$$\Rightarrow 0 = 3mx + x + 3m^2x \Rightarrow 0 = (3m^2+3m+1)x$$

Compare coefficients: $3m^2 + 3m + 1 = 0$. No invariant lines $\Rightarrow b^2 - 4ac < 0$.

$$\therefore 3^2 - 4(3)(1) = -3 < 0$$

So there are no real roots, and hence there are no invariant lines.